

Artificial Intelligence

CS470-GR1 Workshop Review

Instructions: Break into teams as determined by your instructor. Each team will be assigned a question or two to discuss and answer as a group. After 25 minutes each group will address the class explaining their answers. Focus on explaining the underlying concepts, comparing algorithmic approaches, and applying these theories to practical scenarios.

Lesson 1 — Intro to Artificial Intelligence

- 1.1 What are the fundamental tasks that modern artificial intelligence systems perform? Provide at least three concrete examples of different AI applications and detail how they process their specific inputs.
- 1.2 What are the key architectural and computational strengths that allow modern AI systems to excel? Conversely, what are their fundamental limitations and edge-case vulnerabilities?
- 1.3 Why is human-in-the-loop evaluation critical for modern AI outputs?

Lesson 2 — Anatomy of AI Systems

- 2.1 What are the four major components that constitute a standard intelligent agent pipeline?
- 2.2 Describe the journey of an input as it is transformed into an output or decision by an AI system.
- 2.3 How do the distinct techniques of search, learning, and inference map onto different problems sets when considering an AI pipeline?

Lesson 3 — Artificial Intelligence Workshop

- 3.1 How do structural constraints, few-shot examples, and chain-of-thought phrasing intentionally alter the output distribution and attention mechanisms of a Large Language Model?
- 3.2 What are the most common failure cases in deployed AI systems, and what causes them?
- 3.3 In what specific ways does the precision and design of a prompt dictate the quality, focus, and alignment of the resulting output?

Lesson 4 — Problem Formulation

- 4.1 What core components formally define a search problem?
- 4.2 Choose a complex real-world scenario (e.g., logistics routing, autonomous navigation) and formulate it strictly as a formal search problem.
- 4.3 Why is the initial choice of problem representation critical to computational feasibility? Discuss the impact of state abstraction.

Lesson 5 — Uninformed Search

- 5.1 Describe the distinct exploration strategies and contour expansions of Depth-First Search (DFS), Breadth-First Search (BFS), and Uniform Cost Search (UCS).
- 5.2 Under what specific environment constraints are DFS, BFS, and UCS considered complete? Which of these algorithms guarantee optimality, and why?
- 5.3 How does the choice of data structure for frontier management (LIFO stack, FIFO queue, priority queue) intrinsically dictate the behavior of an uninformed search algorithm?

Lesson 6 — Informed Search

- 6.1 What is a heuristic function, and how does it inject domain-specific knowledge to improve search efficiency over uninformed methods?
- 6.2 Compare and contrast Greedy Best-First Search with A* Search in terms of their core evaluation functions and their respective susceptibilities to local obstacles.
- 6.3 How does A* mathematically combine the actual cost-so-far with the estimated remaining cost? Why does this specific combination prevent the algorithm from prematurely accepting a suboptimal goal?

Lesson 7 — Heuristics

- 7.1 What are the formal mathematical definitions of heuristic admissibility and heuristic consistency (monotonicity)?
- 7.2 How is the quality of a heuristic measured? What does it mean practically when one heuristic dominates another?
- 7.3 Explain the theoretical guarantees of A*. Why is it optimal for Tree Search when the heuristic is merely admissible, but requires consistency to be optimal in Graph Search?

Lesson 8 — Adversarial and Stochastic Decision-Making

- 8.1 How does the Minimax algorithm recursively evaluate game trees to make decisions in zero-sum, competitive environments?
- 8.2 What is the fundamental distinction between a static plan and an AI policy? Why are comprehensive policies absolute necessities in stochastic environments?
- 8.3 How does the Expectimax algorithm modify the traditional adversarial search tree to account for unknown action outcomes and probabilistic environmental shifts?
- 8.4 Contrast adversarial reasoning with probabilistic reasoning. Provide an example of a specific real-world domain where each would be the most appropriate approach.